Finkraft B2B E-Invoicing Platform: Master CTO Submission

UAE Federal Tax Authority (FTA) Peppol 5-Corner DCTCE Compliance Platform

Candidate: Srinivas Senniappan | Submission Deliverable: Execution Plan

---

Part 3: 30-Day CTO Execution Plan

1. Executive Strategy & Squad Allocation

With 3 Backend Engineers (BE1, BE2, BE3) and 1 Frontend Engineer (FE1), we operate as a single-threaded squad focused strictly on the compliant critical path. Weeks 1–4 execute feature delivery and achieve full feature freeze; Weeks 5–6 are strictly reserved for partner UAT, security penetration testing, and zero-defect launch hardening.

2. Weekly Execution Schedule (Weeks 1 to 4)

Week 1: Architectural Alignment, Scope Freeze & Core Foundations

* CTO: Enforce scope freeze with CEO/Product; audit ASP sandboxes and rate limits; establish trunk CI/CD with ephemeral previews.
* BE1 (State Machine & Locks): Core state machine, PostgreSQL schema, unique index on (seller_trn, invoice_number) with row locks.
* BE2 (ASP Adapter): Unified ASPAdapter contract for primary launch ASP (A/B), isolating authentication and rate-limiting queues.
* BE3 (PINT AE Validation): PINT AE business validation engine (15-digit TRN validation, BT-48 VAT category codes, tax arithmetic).
* FE1 (Ingestion UI): Authenticated drag-and-drop CSV/Excel bulk upload interface with instant client-side format checks.
* Milestone: Test suite passing with 100% coverage on state transitions, validation errors, and duplicate rejection.

Week 2: The 'Walking Skeleton' (End-to-End Ingestion to Clearance)

* CTO: Drive end-to-end integration across components; establish structured JSON logging with correlation IDs (seller_trn:invoice_number).
* BE1 + BE2: Wire Ingestion -> PINT AE -> ASP Adapter. Implement async worker queue and webhook ingress with HMAC verification.
* BE3: Build Active Reconciliation Worker for submission timeouts and the idempotent webhook event ledger.
* FE1: Invoice Status Dashboard (search by invoice number, state badges: ACKNOWLEDGED, DELIVERED_C3, REPORTED_C5, REJECTED).
* Milestone: First real test invoice uploaded via UI, submitted to ASP sandbox, and tracked to C5 clearance in staging.

Weeks 3–4: Failure Resilience, Bulk Scaling & Feature Freeze

* BE1 (Bulk Processing): Streaming CSV parser (10k+ rows) without memory spikes; tune token-bucket throttler (10 req/s for ASP A).
* BE2 (Secondary ASP & Failover): Finalize secondary adapter in registry to enable customer ASP switching; build Dead Letter Queues.
* BE3 (Observability & Audit): Implement tamper-evident audit logging for FTA inspection; instrument Prometheus metrics and PagerDuty.
* FE1 (Error Triage UI): Row-level error remediation view (download invalid rows with BT-xx error hints so users fix and re-upload).
* CTO + Squad: Chaos Testing Game Day (inject ASP 504 timeouts, duplicate uploads, out-of-order webhooks). FEATURE FREEZE END W4.

3. Deliberate Scope Cuts (What We Cut and Why)

1. Native ERP Connectors (SAP, Oracle, Dynamics): Cut due to multi-month integration cycles. Replaced by standardized CSV/Excel uploads and REST API.
2. Active-Active Multi-ASP Dynamic Load Balancing: Cut to eliminate distributed concurrency races and double-filing risks. Replaced by single ASP per seller with config switching.
3. In-Browser Invoice Editor & Line-Item Correction: Cut because interactive UI tax calculators would consume 100% of FE1's runway. Replaced by 'Reject & Download Error CSV'.
4. Buyer Portal & Advanced Financial Analytics: Cut because compliance is driven by Seller (C1) and FTA (C5). Buyer portals add zero regulatory value for Day 1.

4. The First Two Things to Measure

Metric 1: Zero-Duplicate Submission Rate & Idempotency Violations (Target: Exactly 0)

* Rationale: Sovereign tax invariant. Submitting twice creates phantom tax liability on customer balance sheets and triggers FTA audits.
* Instrumentation: Counter finkraft_duplicates_blocked_total at edge; immediate P0 alert if any duplicate payload reaches an ASP.

Metric 2: End-to-End Compliance Latency (P95 < 120s) & Ambiguous State Dwell Time (< 60s)

* Rationale: Invoices stuck in intermediate or ambiguous states delay commercial payments. Verifies reconciliation auto-heals glitches.